

Poisson–Charlier Spectrum and Trace Class for the M/M/∞ Flow

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Abstract

We give a convention-complete theorem for the infinite-server immigration–death chain. The unique positive-rate invariant law is Poisson, and the transition kernel is exactly a binomial survivor count convolved with a Poisson immigration count. Charlier polynomials diagonalize the generator on L^2 of that law, yielding the sharp gap, all relaxation rates, and a positive-time trace-class semigroup with an explicit source Fredholm product. Coupling gives quantitative total-variation bounds. Pure-death, pure-birth, zero-rate, small-intensity and long-time faces are kept separate. The source semigroup is not relabeled as an arithmetic primitive-orbit determinant. Independent exact, symbolic, replay and hostile-mutation audits accompany the proof.

1 Generator and exact kernel

For $n \in \mathbb{N}_0$ let

$$Qf(n) = \lambda\{f(n+1) - f(n)\} + \mu n\{f(n-1) - f(n)\}, \quad \lambda, \mu > 0. \quad (1)$$

Write $\rho = \lambda/\mu$ and $a_t = e^{-\mu t}$. The physical clock is the continuous-time Markov time t .

Theorem 1 (Poisson–Charlier atlas). *The unique reversible invariant law of (1) is*

$$\pi_n = e^{-\rho} \frac{\rho^n}{n!}, \quad n \geq 0.$$

Conditionally on $X_0 = n$, one has the exact decomposition

$$X_t \stackrel{d}{=} B_{n,a_t} + P_{\rho(1-a_t)}, \quad (2)$$

where the summands are independent, B_{n,a_t} is binomial and P_m is Poisson with mean m . Equivalently,

$$\mathbb{E}[z^{X_t} \mid X_0 = n] = ((1 - a_t) + a_t z)^n e^{\rho(1-a_t)(z-1)}. \quad (3)$$

Define C_k by

$$\sum_{k \geq 0} C_k(n; \rho) \frac{u^k}{k!} = e^{-\rho u} (1 + u)^n. \quad (4)$$

Then $\mathbb{E}_\pi[C_k C_\ell] = k! \rho^k \mathbf{1}_{k=\ell}$ and, for the normalized $\phi_k = C_k / \sqrt{k! \rho^k}$,

$$Q\phi_k = -k\mu\phi_k, \quad P_t\phi_k = e^{-k\mu t}\phi_k. \quad (5)$$

Thus the $L^2(\pi)$ gap is μ and, for $t > 0$,

$$\mathrm{Tr} P_t = \frac{1}{1 - e^{-\mu t}}, \quad \det(I - zP_t) = \prod_{k \geq 0} (1 - ze^{-k\mu t}) \quad (6)$$

in the source-local trace/Fredholm convention.

Proof. Detailed balance is $\lambda\pi_n = \mu(n+1)\pi_{n+1}$, which gives the Poisson law and reversibility. Independently thinning each initial customer and superposing the stationary immigration process gives (2), hence (3). Taking coefficients in (4) gives the displayed Charlier polynomial. Applying (1) to that polynomial yields $QC_k = -k\mu C_k$; Poisson coefficient orthogonality gives the stated norm and completeness. The spectral expansion then proves (5). Summing the geometric eigenvalue sequence gives the trace in (6), and absolute convergence of $\sum_k |ze^{-k\mu t}|$ gives the source Fredholm product. $\square$

2 Mixing and boundary faces

A shared-immigration coupling leaves each unmatched initial customer alive with probability a_t . Therefore

$$\|P_t(n, \cdot) - P_t(m, \cdot)\|_{\text{TV}} \leq \min\{1, a_t|n - m|\}. \quad (7)$$

Coupling a deterministic initial state to a stationary Poisson initial state also gives $\|P_t(n, \cdot) - \pi\|_{\text{TV}} \leq \min\{1, a_t(n + \rho)\}$. These are deliberately coarse but uniform bounds; the kernel ledger records the finite PMF window and its explicit tail mass.

If $\lambda = 0$ and $\mu > 0$, the pure-death chain has the absorbing law δ_0 . If $\mu = 0$ and $\lambda > 0$, pure birth has no stationary probability and escapes to infinity. If both rates vanish, the identity chain has every point mass stationary. As $\rho \downarrow 0$ the Poisson law collapses weakly to δ_0 ; as $\mu \downarrow 0$ with λ fixed it loses tightness. For positive rates, $a_t \downarrow 0$ proves total-variation convergence to π .

Source note and declarations

The birth–death spectral framework follows Karlin and McGregor [1]; the Poisson polynomial normalization follows the Charlier construction [2]. The queueing interpretation is standard [3]. No target prime or zero table, local arithmetic datum, Euler factor, root number, automorphy object, target functional equation, Hilbert–Polya operator or Route-B input is used. Generative tools assisted drafting and code generation; the artifact chain checks the displayed claims. Scope literal: NO_BAD_EULER_OR_ROOT_NUMBER.

References

- [1] S. Karlin and J. McGregor, “The differential equations of birth-and-death processes, and the Stieltjes moment problem,” *Transactions of the American Mathematical Society* 85 (1957), 489–546. DOI: 10.1090/S0002-9947-1957-0091566-1.
- [2] C. V. L. Charlier, “Über die Darstellung willkürlicher Funktionen,” *Arkiv för Matematik, Astronomi och Fysik* 16 (1929).
- [3] W. A. Massey, *An Introduction to Queueing Networks*, Springer, 1987. DOI: 10.1007/978-1-4612-4960-1.